

Euclidean Vector Spaces

1. Vectors and Euclidean Space

Definition

- A **vector** in $\mathbb{R}^n$ is an ordered list of n real numbers, written as:

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}, \quad v_i \in \mathbb{R}.$$

- The **Euclidean n -space** ($\mathbb{R}^n$) is the set of all such vectors.
- Geometric Interpretation:** A vector can represent a point in space or an arrow from the origin to that point.

Example

- In $\mathbb{R}^3$, $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ represents the point $(1,2,3)$ or an arrow from $(0,0,0)$ to $(1,2,3)$.

2. Vector Algebra

Operations

1. Addition:

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}.$$

2. Scalar Multiplication:

$$c\mathbf{u} = \begin{pmatrix} cu_1 \\ cu_2 \\ \vdots \\ cu_n \end{pmatrix}, \quad c \in \mathbb{R}.$$

Properties

- Commutative:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- Associative:** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- Zero Vector:** $\mathbf{0} + \mathbf{v} = \mathbf{v}$.
- Distributive:** $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
- Scalar multiplication:** $a\mathbf{v} \in \mathbb{R}^n$.
- Closure under addition:** $\mathbf{u} + \mathbf{v} \in \mathbb{R}^n$.

MATLAB Code

```
u = [1; 2; 3]; % Column vector
v = [4; 5; 6];
sum = u + v;   % Vector addition
scaled = 3 * u; % Scalar multiplication
```

3. Dot Product, Norm, and Distance

Dot Product

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n.$$

Dot Product (Inner Product)

- For $\mathbf{u} = (u_i), \mathbf{v} = (v_i) \in \mathbb{R}^n$:

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i.$$

Norm (Length)

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}.$$

- Unit vector:** $\|\mathbf{u}\| = 1$.
To normalize: $\frac{\mathbf{u}}{\|\mathbf{u}\|}$.

Distance Between Vectors

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + \cdots + (u_n - v_n)^2}.$$

- Distance:** $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

Properties:

- Symmetry:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
- Linearity:** $\mathbf{u} \cdot (a\mathbf{v} + b\mathbf{w}) = a(\mathbf{u} \cdot \mathbf{v}) + b(\mathbf{u} \cdot \mathbf{w})$.
- Positive definiteness:** $\mathbf{u} \cdot \mathbf{u} \geq 0$, equality iff $\mathbf{u} = \mathbf{0}$.

Example

- $\mathbf{u} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$:

$$\mathbf{u} \cdot \mathbf{v} = (1)(3) + (-1)(0) + (2)(1) = 5,$$

$$\|\mathbf{u}\| = \sqrt{1^2 + (-1)^2 + 2^2} = \sqrt{6},$$

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{(1-3)^2 + (-1-0)^2 + (2-1)^2} = \sqrt{6}.$$

MATLAB Code

```
u = [1; -1; 2];  
v = [3; 0; 1];  
dot_product = dot(u, v); % Dot product  
norm_u = norm(u); % Norm of u  
distance = norm(u - v); % Distance between u and v
```

4. Linear Combinations and Spans

Definitions

- A **linear combination** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k, \quad c_i \in \mathbb{R}.$$

- The **span** of $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of all their linear combinations.

Algorithm to Check Span

To determine if $\mathbf{b} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$:

1. Form the matrix $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_k]$.
2. Solve $A\mathbf{x} = \mathbf{b}$.
 - If consistent, $\mathbf{b}$ is in the span.
 - If inconsistent, $\mathbf{b}$ is not in the span.

Example

Check if $\mathbf{b} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$ is in $\text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix}\right\}$:

1. Solve $c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$.
2. System:

$$\begin{cases} c_1 + 2c_2 = 5 \\ c_1 + 3c_2 = 5 \end{cases} \Rightarrow c_1 = 5, c_2 = 0.$$

Thus, $\mathbf{b}$ is in the span.

MATLAB Code

```
A = [1 2; 1 3]; % Columns are v1 and v2
b = [5; 5];
x = A \ b;      % Solve Ax = b (if possible)
```

5. Subspaces

Definition

A subset $V \subseteq \mathbb{R}^n$ is a **subspace** if:

1. $\mathbf{0} \in V$.
2. Closed under addition: $\mathbf{u}, \mathbf{v} \in V \Rightarrow \mathbf{u} + \mathbf{v} \in V$.
3. Closed under scalar multiplication: $\mathbf{u} \in V, c \in \mathbb{R} \Rightarrow c\mathbf{u} \in V$.

Key Examples

1. **Solution Space of Homogeneous Systems:**
 $V = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$ is a subspace.
2. **Span of Vectors:** $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a subspace.

Example

- Let $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + y - z = 0 \right\}$.
 - **Verification:**
 - $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \in V$.

- If $\mathbf{u}, \mathbf{v} \in V$, then $(u_1 + v_1) + (u_2 + v_2) - (u_3 + v_3) = 0 \Rightarrow \mathbf{u} + \mathbf{v} \in V$.
- Closed under scalar multiplication similarly.

6. Linear Independence

Definition

Vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ are **linearly independent** if:

$$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k = \mathbf{0} \Rightarrow c_1 = \dots = c_k = 0.$$

Otherwise, they are **linearly dependent**.

Algorithm to Check Independence

1. Form the matrix $A = [\mathbf{v}_1 \dots \mathbf{v}_k]$.
2. Compute $\text{rref}(A)$.
 - If all columns are pivot columns $\Rightarrow$ linearly independent.
 - If there's a non-pivot column $\Rightarrow$ linearly dependent.

Example

Check if $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}$ are linearly independent:

1. $A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$.
2. Non-pivot column exists $\Rightarrow$ linearly dependent.

MATLAB Code

```
A = [1 2; 1 2];
rref_A = rref(A); % Result: [1 2; 0 0]
```

7. Basis and Coordinates

Basis

A set $B = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a **basis** for a subspace V if:

1. B is linearly independent.
2. $\text{span}(B) = V$.

Coordinates

If B is a basis for V , any $\mathbf{v} \in V$ can be written uniquely as:

$$\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k.$$

The **coordinates** of $\mathbf{v}$ relative to B are:

$$[\mathbf{v}]_B = \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix}.$$

Example

Let $B = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$ be a basis for $\mathbb{R}^2$. Find $[\mathbf{v}]_B$ for $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$:

1. Solve $c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.
2. System:

$$\begin{cases} c_1 + c_2 = 3 \\ c_1 - c_2 = 1 \end{cases} \Rightarrow c_1 = 2, c_2 = 1.$$

Thus, $[\mathbf{v}]_B = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

MATLAB Code

```
B = [1 1; 1 -1]; % Basis vectors as columns
v = [3; 1];
coordinates = B \ v; % Solve B * x = v
```

8. Dimension

Definition

- The **dimension** of a subspace V is the number of vectors in any basis for V .
- $\dim(\mathbb{R}^n) = n$, $\dim(\{\mathbf{0}\}) = 0$.

Key Theorems

1. If $\dim(V) = k$, any set with $> k$ vectors in V is linearly dependent.
2. If $\dim(V) = k$, any set with $< k$ vectors cannot span V .

Determinant via Cofactor Expansion

For a 3×3 matrix:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

Area of a Triangle

For vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\text{Area} = \frac{1}{2} \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}.$$

Example

- Let $V = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$.
 - A basis for V has 2 vectors $\Rightarrow \dim(V) = 2$.

9 Equivalent Statements for Invertibility

For an $n \times n$ matrix A , the following are equivalent:

1. A is invertible.
2. $\det(A) \neq 0$.
3. The RREF of A is I_n .

- 4. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- 5. The columns of A form a basis for $\mathbb{R}^n$.

MATLAB Commands

Command	Purpose
<code>dot(u, v)</code>	Compute dot product of vectors.
<code>norm(u)</code>	Compute norm of a vector.
<code>A \ b</code>	Solve linear system $A\mathbf{x} = \mathbf{b}$.
<code>rref(A)</code>	Compute reduced row-echelon form.
<code>B \ v</code>	Find coordinates of $\mathbf{v}$ in basis $\mathbf{B}$.